TEST PLAN

1. Testing Objectives and Scope

Testing Objectives

The main objective of this testing is to check whether the API correctly returns the second largest number from the array provided in the request. The testing also ensures that the API response correctly when different types of input data are provided.

Another objective is to check how the API handles **invalid inputs, edge cases, and unauthorized requests**.

Scope of Testing

The scope of testing includes:

- Verifying correct functionality for valid inputs.
- Testing different types of arrays such as random numbers, duplicates, negative numbers, and decimal values.
- Testing edge cases such as arrays with two elements, empty arrays, and large arrays.
- Performing negative testing with invalid inputs such as null values, boolean values, and string values.
- Testing security scenarios such as unauthorized access and invalid tokens.

2. Test Strategy and Approach

The API testing was performed using **Postman**.

The following approach was used during testing:

1. Configure the API endpoint in Postman.
2. Add the required **Bearer Token authorization**.
3. Send requests using the **POST method**.
4. Provide different input arrays in the request body.
5. Verify whether the API returns the correct second largest number.
6. Perform negative testing by sending invalid inputs.
7. Capture Postman screenshots and document the test results.

Both valid and invalid test scenarios were executed to check how the API behaves in different situations.

3. Resources (People, Tools, Environment)

People

- Tester: Kunal Kadam

Tools

- Postman (API Testing Tool)
- Microsoft Excel (for test case documentation)
- Microsoft Word / Google Docs (for test plan documentation)

Test Environment

- API Endpoint: http://ec2-3-85-189-88.compute-1.amazonaws.com:3000/second-largest
- HTTP Method: POST
- Data Format: JSON

4. Schedule and Milestones

Activity	Description
Test Planning	Understanding the API requirements and preparing the test plan
Test Case Preparation	Creating functional and negative test cases
Test Execution	Running API requests using Postman
Test Documentation	Recording test results and capturing screenshots
Final Submission	Preparing test plan, screenshots, and test case documents

5. Entry and Exit Criteria

Entry Criteria

Testing will start when:

- The API endpoint is accessible.
- Postman tool is configured.
- Authorization token is available.
- Test cases are prepared.

Exit Criteria

Testing will be considered complete when:

- All test cases are executed.
- Test results are documented.
- Any issues or observations are recorded.
- All deliverables are prepared for submission.

6. Risks and Mitigation

Risk	Mitigation
API server unavailable	Retry testing after server is available
Invalid or missing documentation	Verify API response through testing
Authorization token issues	Use valid token for authentication
Unexpected API response	Document the observation in test results

7. Deliverables and Responsibilities

Deliverables

The following documents will be submitted as part of the assignment:

1. **Test Plan Document (PDF)**
2. **Test Cases Document (Excel)**
3. **Postman Screenshots showing API testing**

Responsibilities

The tester is responsible for:

- Preparing the test plan.
- Designing and executing test cases.
- Recording test results.
- Identifying and documenting any issues found during testing.

8. Defects / Issues Observed During Testing

During testing, a few issues were observed in the API response:

• The API accepts invalid input values such as null and boolean values instead of rejecting them. The API should validate the input properly and allow only numeric values in the array.

• The API returned an incorrect second largest value when mixed positive and negative numbers were provided, indicating a possible issue in the calculation logic.

• The API accepts special characters and malicious string inputs instead of rejecting them, which indicates missing input validation and potential security risks.